WORKING PAPER 01 · METHODOLOGY AND DATA CRITIQUE

The Substance of an Economy

Measuring reproduction rather than output: a framework for asking where a capability came from, and whether it would survive.

Output metrics answer one question well: how much market activity occurred in a territory during a period. They answer a different question badly, and that second question is the one that decides whether a society holds together. This paper sets out an alternative object of measurement: the stock of capability a society actually holds, and the process by which it reproduces that stock.

The framework adds a test that existing indices do not apply. For each domain of capability it asks not only how much is present, but whether it was produced at home or imported, and whether it would survive the withdrawal of its current source. Capability that is purchased, subsidised or staged scores high on level and low on substance. The paper specifies the indicator set, the aggregation algebra, a non-compensatory scoring rule, a robustness protocol, and a ledger of what each underlying data source can and cannot bear.

Document type	Working paper: methodology and data critique
Audience	Research, statistical and policy-analysis audiences
Geographic scope	Framework is jurisdiction-agnostic; examples drawn from national and sub-national cases
Data status	No country scores are published here. The demonstration dataset is synthetic and is labelled as such throughout.
Version	1.0

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A note on what is not here

This paper deliberately publishes no country rankings. The contribution is the instrument, not a scoreboard. Every number in the demonstration exhibits comes from a synthetic dataset of ten archetypal profiles, built to show how the algebra behaves. Populating the instrument with real indicator values is a separate exercise and is set out as the research agenda in section 9.

1 The measurement problem

An economy that grows while its people become less capable of acting in the world is not a growing economy. It is an economy consuming its own foundations. The distinction is invisible in the statistic that most public argument relies on.

This is not a complaint about GDP's accuracy. GDP is a well-constructed measure of market production on a territory during a period, and for that purpose it is close to the best instrument available. The problem is that a flow measure has been assigned a stock question, and the assignment is so routine that the mismatch is no longer noticed.

1.1 Four structural biases, not four errors

The shortcomings are systematic. Each follows from how the accounts are built, and each therefore pushes in a predictable direction rather than adding noise.

1. **It counts only what is priced.** Non-market production sits outside the boundary. India's 2019 Time Use Survey found women spending 305 minutes a day on unpaid work against 67 minutes for men; of the women's total, 281 minutes fall in non-SNA production, activity the accounts do not see at all. A country can raise measured output by moving care work into the market without any household becoming better off.
2. **It is gross, not net.** Depletion of natural capital and depreciation of human and institutional capital are not deducted. Output extracted by running down a stock is recorded as income.

3 **It is territorial, not residential.** Value added booked in a jurisdiction need not accrue to the people who live there. The gap between what is produced on a territory and what is received by its residents is the space in which several of the paradoxes in section 1.3 live.

4 **It is indifferent to composition.** Defensive expenditure adds to the total. Treating illness caused by air pollution, cleaning up a spill, commuting further to reach work all register as growth.

The Commission on the Measurement of Economic Performance and Social Progress put the conclusion plainly in 2009: GDP is not wrong as such, but wrongly used. Its first recommendation was that material well-being be assessed through income and consumption rather than production. Seventeen years later the recommendation remains a minority practice in public argument.

1.2 Why the gap has widened rather than narrowed

Three changes in the structure of production made the mismatch worse after the commission reported. Intangible capital became mobile, so the location of value added became a legal choice rather than a physical fact. Global value chains separated the site of production from the site of booking, with contract manufacturing recording output in the jurisdiction that owns the blueprint rather than the one that operates the factory. Household balance sheets became the transmission channel for growth, so measured output could expand while household net worth fell.

1.3 Two artefacts worth keeping in view

These are not curiosities to be adjusted away in a footnote. They are the ordinary behaviour of a flow metric applied to a stock question, and they appear whenever the conditions are right.

+26.3%

IRELAND, 2015 REAL GDP

Against +6.4% growth in net national income the same year.

145 / 189

EQUATORIAL GUINEA HDI RANK

With the highest GDP per capita in Africa at the time.

70%

LEARNING POVERTY, LOW- AND MIDDLE-INCOME COUNTRIES

Share of 10-year-olds unable to read and understand a simple text, post-pandemic.

55%

SDG INDICATORS WITH ANY REPORTED VALUE

Average across countries, 2015-2019. Twenty-two countries reported under a quarter.

Ireland's Central Statistics Office reported 26.3% real GDP growth for 2015 after a revision driven by multinationals relocating intellectual property into the jurisdiction. Gross national income rose 18.7%; net national income rose 6.4%. Paul Krugman called it leprechaun economics. The Irish Central Bank subsequently abandoned GDP and GNP as its headline measures and constructed a modified gross national income series instead. A statistical agency concluded that its own core aggregates no longer described its economy.

Equatorial Guinea presents the same mismatch in a different mechanism. Oil rents delivered the highest GDP per capita on the continent, while the country ranked 145th of 189 on the Human Development Index, with an estimated poverty rate near 77%. In 2011 it showed the largest gap of any country between its rank on gross national income per capita and its rank on human development: 91 places. The rent did not need the population, so the state had no fiscal reason to invest in it. That is a governance outcome, but it is an outcome the output statistic concealed for two decades.

1.4 The claim

The object worth measuring is the stock of capability and the process that reproduces it. The diagnostic question that existing instruments do not ask is one of provenance: where did this capability come from, and would it survive the withdrawal of its current source? A health system staffed by clinicians who can be repatriated, a school system whose results depend on household spending on tutoring, a pension that exists because a government chose to fund it this year, all report high capability and hold very little.

2 What already exists, and where it stops

The critique of GDP is well developed and the alternative instruments are numerous. The gap this paper addresses is narrower and more specific than the general complaint: the existing family of composite indices measures how much capability is present at a moment, and none of them measures whether that capability is self-reproducing.

2.1 Capabilities, and what the HDI bought

Sen's argument was that development is the expansion of freedoms rather than the accumulation of commodity command, and that the relevant space is what people are able to do and to be. The Human Development Index operationalised a thin version of that: health, knowledge, income, aggregated since 2010 as a geometric mean. Its virtues are parsimony and a quarter-century of institutional legitimacy. Its limits are dimensional coverage and the fact that each dimension is measured at the level of the average, which makes distribution invisible.

2.2 After 2009: a settlement, not a solution

The commission's twelve recommendations produced a durable intellectual settlement: no single measure can summarise something as complex as a society, so build a dashboard. That settlement was correct and it has been productive. It also moved the argument away from the question of what to measure and towards the question of how many panels the dashboard should have, which is a less important question.

2.3 The composite-index family compared

Six instruments illustrate the range. Each is competently built; the column that matters is the last one.

Instrument	Dimensions	Aggregation	Captures	Cannot capture
Human Development Index	3: health, education, income	Geometric mean of 3	Average capability in three basic dimensions	Distribution; institutional quality; whether capability is produced or purchased
Social Progress Index	3 dimensions, 12 components, 57 indicators	PCA weights within component; geometric across	Outcomes in basic needs, wellbeing and opportunity, explicitly independent of income	Provenance and durability by design: it measures outcomes, not inputs, and so cannot see what sustains the outcome
OECD Better Life Index	11 topics	User-weighted average	Multi-dimensional current wellbeing plus a comparison of what citizens prioritise	Cross-country comparability of weights; sustainability; origin of capability
Global MPI (OPHI)	3 dimensions, 10 indicators	Counting approach with dual cut-off	Joint deprivation at household level, with dimensional breakdown	Non-poor capability; system-level reproduction; the middle of the distribution
Worldwide Governance Indicators	6 dimensions	Unobserved components model on perception sources	Expert and survey perceptions of governance	Direct observation; it is perception of institutions, weighted towards the views of a narrow panel
World Happiness Report	Life evaluation plus 6 correlates	Cross-sectional regression	Subjective life evaluation and its statistical correlates	Any causal or stock interpretation; subject to reference-group and language effects

Table 1. The composite-index family, and the shared blind spot in the final column.

2.4 The blind spot is provenance, and it is structural

The Social Progress Index made an explicit and defensible design choice: measure outcomes, not inputs, so that the index is independent of income and of government effort. That choice delivers exactly what it was meant to deliver. It also means the index cannot distinguish an outcome that a society produces from one that it purchases. A country importing its clinicians and its medicines, and financing them from a depleting rent, scores identically to a country that trains and manufactures its own, for as long as the money lasts.

The critique of the SPI in the measurement literature has been about interpretability and about the trade-offs implied by endogenous weights, not about provenance. That is the right critique of that instrument, and it leaves the gap open.

A second structural problem is compensability. Additive aggregation allows a surplus in one dimension to offset a deficit in another, which is defensible for a pure economic index and indefensible here. The OECD and JRC handbook is explicit: where the analyst judges that a gain in one domain cannot compensate for a loss in another, neither linear nor geometric aggregation is suitable and a non-compensatory approach is required. Section 5 takes that instruction seriously rather than treating it as a caveat.

The gap, stated precisely

Existing indices answer: how much capability is present? The missing question is: is that capability endogenous, and is it durable? A framework that asks all three turns a scoreboard into a diagnostic, and it is the diagnostic that has policy content.

3 The framework

3.1 From flows to stocks

The reframing is a change of accounting object. A society holds stocks of capability: the competence of its adults, the health of its people, the productive capacity of its workers, the security of its households, the administrative reach of its state, the material command of its households. Those stocks generate a flow of functioning each year. Measured output is closer to the flow than to the stock.

Once the object is a stock, the possibility of consuming it becomes visible. A country can raise this year's flow by drawing down the stock: by working children rather than educating them, by exporting clinicians rather than retaining them, by running down a water table rather than charging for water, by financing consumption from a depleting rent rather than from production. Every one of these raises measured output now and lowers capability later, and no flow statistic distinguishes the two cases.

3.2 Three ways a capability can be false

The framework's central distinction is between capability that is present and capability that is self-reproducing. Three failure modes recur, and each has a distinct signature in the data.

- **Imported.** The capability is present but produced elsewhere. Health systems staffed by expatriate clinicians; essential medicines whose active ingredients are all imported; schools following a foreign examination board. The level is high; the capability leaves when the contract does.
- **Subsidised.** The capability exists only while a transfer continues: a subsidy, a donor programme, a commodity rent, a public payroll that is really an employment scheme. Withdraw the transfer and the capability goes with it.
- **Staged.** The capability exists as an event rather than a system. A health camp, a saturation drive, a launch with a ribbon. Reported reach is high and the institutional base is thin, so the effect decays between campaigns and cannot be audited afterwards.

These are not moral categories and none of them is a criticism of the people delivering the service. They are statements about what would happen to the capability under a plausible change in conditions, and that is a factual question with observable proxies.

3.3 The LED test

For each domain, three questions are scored independently.

Axis	Question	Illustrative proxies
L — Level	Is the capability adequate, measured by direct assessment rather than by proxy?	Learning poverty; effective health coverage; informal employment share; catastrophic health spending
E — Endogeneity	Is the capability produced within the domestic institutional economy, or imported, rented or donor-supplied?	Import dependence of medicines; expatriate share of clinicians; remittance and rent share of income; external share of health financing
D — Durability	Would the capability survive a plausible shock: a funding withdrawal, a political transition, a price spike, a programme being renamed?	Statutory versus scheme-based entitlement; retention through a consolidation year; learning retained after school closure; revenue retained after a commodity fall

Table 2. The three axes. Level is the conventional question; Endogeneity and Durability are what the existing instruments leave out.

3.4 Formal statement

For domain d and jurisdiction i, let L, E and D be normalised to the unit interval. Domain substance is the weighted geometric mean

$$S(i,d) = L(i,d)^{0.50} \times E(i,d)^{0.25} \times D(i,d)^{0.25}$$

The geometric form is chosen because the axes are not substitutes. A domain that is fully present but entirely imported is not two-thirds of a capability; it is a contingent one, and the product form expresses that. Level carries half the weight because it is the only axis measured directly rather than by proxy; giving the proxies equal weight would let measurement uncertainty dominate the result.

Domain scores aggregate to a jurisdiction score with a binding-constraint penalty:

$$SI(i) = G(i) \cdot [(1 - \gamma) + \gamma \cdot \min_d S(i,d)]$$

where G is the geometric mean across domains and gamma in [0,1] sets how far the weakest domain drags the total. At gamma = 0 the score is a pure geometric mean, which is the HDI's rule. At gamma = 1 a jurisdiction with a failed domain scores zero regardless of everything else. The default is gamma = 0.5, and section 5.6 requires the whole result to be reported across gamma from 0 to 1 rather than at a single value.

3.5 What the framework refuses to do

- It does not publish a single ranked league table as its product. The product is a profile with a named binding constraint and named fragility flags.
- It does not monetise non-market production or ecosystem services. Those are real and material; the valuation methods are contested enough that folding them in would import more disagreement than information.
- It does not claim cardinal comparability across jurisdictions. The scores are ordinal within a normalisation, and the sensitivity protocol in section 5.6 exists because of that.
- It does not treat any indicator as a target. Section 6.5 sets out why an indicator used for targeting stops measuring the thing it was chosen for.

4 The six domains

Six domains cover the capabilities a society must reproduce to remain a society. Each is stated with the conventional proxy, why that proxy fails, the substance indicators, and the known defect in the underlying data. A seventh domain, ecological and physical reproduction, is deliberately deferred and the reason is given at the end of the section.

4.1 Foundational competence

Can an ordinary adult read, count and reason well enough to act in the world?

Conventional proxy. Mean years of schooling; gross enrolment; university rankings

Why it fails. Years of schooling count time served, not capability acquired. Enrolment measures presence in a building. Neither detects a child who has sat through eight grades without learning to read a paragraph.

Substance indicators.

- Learning poverty: share of 10-year-olds below minimum reading proficiency
- Functional numeracy at age 15 on an applied task (unitary method, dosage, interest)
- Adult literacy measured by direct assessment, not self-report
- Unsubsidised learning rate: share at proficiency without paid tutoring
- Teacher time-on-task from direct classroom observation

Endogeneity proxies. Share of teachers trained in domestically financed institutions; Curriculum and examination authority held domestically; Reliance on foreign school boards or imported textbooks.

Durability proxies. Learning retained after a six-month interruption in instruction; Share of proficiency that survives the withdrawal of household tuition spending; Assessment cadence: is the country measuring itself, or waiting for a donor survey?.

Known data defect. International assessments are voluntary, so the countries that participate are not a random sample. Learning poverty imputes regional values for non-reporting countries. Citizen-led surveys such as ASER are rural-only and district-sampled.

4.2 Health capability

Does illness impoverish a household, and would care survive without the current programme?

Conventional proxy. Life expectancy; health spending as a share of GDP; hospital beds per 1,000

Why it fails. Life expectancy is a lagging stock that moves slowly and hides distribution. Spending measures input, not access. Beds measure capacity to admit, not capacity to cure, and say nothing about who pays.

Substance indicators.

- UHC service coverage index (tracer services, geometric mean across four categories)
- Catastrophic out-of-pocket spending: share of households above 10% of budget
- Impoverishing health spending: share pushed below the poverty line by medical bills
- Effective coverage: hypertension controlled, not merely diagnosed
- Public-sector health worker density and rural vacancy rate
- Domestic share of essential medicine and diagnostic production

Endogeneity proxies. Active pharmaceutical ingredient and device import dependence; Reliance on foreign-trained clinicians and donor-financed vertical programmes; External share of current health expenditure.

Durability proxies. Is entitlement statutory, or does it lapse with a named scheme?; Coverage retained through a fiscal consolidation year; Primary care staffing retained through a political transition.

Known data defect. UHC coverage is a modelled index of tracer indicators, not an observation of care received. Out-of-pocket shares are survey-based and understate forgone care: a household that cannot afford treatment never appears in the spending data.

4.3 Work that sustains a life

Does a job reproduce a household, or merely register as activity?

Conventional proxy. Unemployment rate; GDP per worker; formal job-creation counts

Why it fails. The unemployment rate excludes anyone who has stopped looking and counts an hour of informal piece work as employment. In economies where most work is informal, it measures almost nothing.

Substance indicators.

- Labour force participation, total and female, age 15 and over
- Informal employment share of total employment
- Share of workers with a written contract, paid leave and pension entitlement
- Labour share of national income
- Real wage growth less productivity growth
- Youth not in employment, education or training (NEET)
- Unpaid care burden in minutes per day, by sex (non-SNA production)

Endogeneity proxies. Remittance share of household income; Commodity rent share of exports; export market concentration; Dependence on a single external buyer or single sector for employment.

Durability proxies. Employment retained through a demand shock; Share of jobs that exist without subsidy, protection or a public payroll; Re-employment speed after displacement.

Known data defect. Labour force surveys change methodology. India moved from the quinquennial Employment-Unemployment Survey to an annual PLFS in 2017-18, and the treatment of unpaid helpers in household enterprises materially shifts the female participation rate. Series breaks are routinely read as trends.

4.4 Security against shocks

If income stops tomorrow, does the household survive it?

Conventional proxy. Law and order; homicide rate; rule-of-law perception indices

Why it fails. Order is the precondition of security, not its content. A state can be orderly and still leave every household one illness away from destitution. Perception indices of governance also track the opinions of a narrow expert panel.

Substance indicators.

- Effective social protection coverage (at least one benefit)
- Pension receipt above statutory retirement age
- Share able to raise emergency funds within 30 days
- Depth of health cover: share of the cost borne, not share of people enrolled
- Moderate or severe food insecurity (FIES)
- Shock responsiveness: coverage expansion sustained beyond the emergency

Endogeneity proxies. Share of protection financed domestically and delivered by statute; Reliance on donor or NGO delivery of core transfers; Whether benefits are contributory entitlements or discretionary transfers.

Durability proxies. Coverage retained after an election; Benefit real value retained through an inflation episode; Portability across employers, sectors and states.

Known data defect. Effective coverage counts receipt of at least one benefit, so a country paying a small child grant to many households scores well while leaving unemployment and disability unprotected. Adequacy is not measured.

4.5 Fiscal and administrative capacity

Can the state tax, count, adjudicate and enforce?

Conventional proxy. Ease of doing business; government effectiveness perception score

Why it fails. Business-friendliness measures the cost of compliance for firms, which is a different object from the capacity of the state to raise revenue, produce statistics and enforce contracts for everyone else.

Substance indicators.

- Tax to GDP, and the direct tax share of total tax
- Tax effort: actual revenue against estimated capacity frontier
- Local government own-source revenue as a share of local spending
- Judicial pendency: stock of cases, and median time to disposal
- Statistical performance: can the state produce its own indicators?
- Capital versus subsidy composition of public spending

Endogeneity proxies. Revenue dependence on resource rents, aid or a single tax handle; Dependence on external technical assistance to run core surveys; Whether the fiscal base is domestic economic activity or an offshore booking centre.

Durability proxies. Revenue retained through a commodity price fall; Institutional authority vested in offices rather than individuals; Continuity of statistical releases across administrations.

Known data defect. Statistical capacity is measured by the World Bank SPI across 51 indicators covering only 14 of 22 framework dimensions; eight dimensions have no established measure at all. The capacity to measure capacity is itself unmeasured.

4.6 Household material substance

Does the output reach the household, and what does it buy?

Conventional proxy. GDP per capita, at purchasing power parity

Why it fails. Output produced on a territory is not income received by its residents. Repatriated profit, depreciation and profit-shifting all sit between the two. A per-capita average also says nothing about the median household.

Substance indicators.

- Real household consumption per capita
- Household final consumption as a share of GDP
- Food share of the household budget (Engel coefficient)
- Housing cost as a share of disposable income
- Household debt service ratio
- Top-decile income share; Gini
- Profit-shifting-adjusted national income where the gap is material

Endogeneity proxies. Remittance share of household consumption; Import dependence of the staple consumption basket; Whether household income derives from domestic production or transferred rent.

Durability proxies. Consumption retained through a currency or terms-of-trade shock; Asset holdings that can be drawn down; Real income retained through a food and fuel price spike.

Known data defect. Household consumption surveys are infrequent and expensive, and their consumption aggregates are not reconciled to national accounts. Survey-to-accounts gaps of large size are common and rarely explained.

4.7 The domain deliberately left out

Ecological and physical reproduction, meaning water, soil, energy reliability, sanitation and natural capital depletion, belongs in any complete account of what sustains a society. It is excluded from the scored set here for two reasons, and both are admissions rather than arguments. First, natural capital valuation methods are contested enough that including them would make the composite's behaviour dominated by a methodological dispute rather than by the phenomenon. Second, the durability axis is already asking a sustainability question about institutions, and running two sustainability logics in one instrument produces double counting that is hard to defend.

The practical recommendation is that ecological reproduction be carried as a separate satellite account reported alongside the composite, on the model of satellite accounts in national statistics, rather than folded into the score.

5 Method

5.1 Normalisation

Indicators are normalised by min-max against fixed goalposts rather than against observed minima and maxima. Observed extremes move when a country has an unusual year, and a moving goalpost makes a series non-comparable to itself over time. Goalposts are set to the theoretical or policy-relevant bound: zero to universal coverage for a coverage indicator, zero to a defined adequacy floor for an inverse indicator such as out-of-pocket share.

The cost of fixed goalposts is that they are a judgement, and they are stated in Appendix A so that they can be disputed. Distance-to-frontier normalisation is the alternative and is preferable when the question is catch-up rather than absolute adequacy; it is not used here because the framework's question is adequacy.

5.2 Weighting

Indicators within a domain carry equal weight. Domain weights are equal. This is a choice made for transparency rather than from a belief that the domains are equally important, and the literature's two alternatives were considered and rejected. Principal components weighting makes the weights endogenous to the sample, which breaks the link between an improvement in one indicator and a movement in the composite, and makes the index difficult to interpret for anyone who did not build it. Benefit-of-the-doubt weighting lets each jurisdiction choose the weights most favourable to itself, which is defensible for a benchmarking exercise and indefensible for a diagnostic one, because it assigns the weakest domain the smallest weight.

5.3 Aggregation, and why compensability is the central question

Aggregation is where a wellbeing index most often goes wrong, and the failure is invisible in the published score. Under any additive rule, a jurisdiction with excellent health and failing foundational competence can outscore one that is adequate everywhere. Whether that is acceptable is a substantive judgement, not a technical one, and this framework's answer is that it is not acceptable: a society whose children cannot read has a binding constraint, whatever else it has achieved.

5.4 The binding-constraint penalty in practice

The penalty term does two jobs. It makes the score sensitive to the weakest domain, which is where policy attention belongs. And it converts the score into a diagnostic, because reporting min_d S(i,d) alongside the composite names the constraint. A jurisdiction that improves its weakest domain gains twice: once through the geometric mean and once through the penalty. That is the intended incentive and it is the principal reason for preferring this rule to a pure geometric mean.

5.5 Missing data

Missing indicators are not imputed. A domain with fewer than three populated indicators is reported as unpopulated and excluded from the geometric mean, with the count of populated indicators shown. The alternative, imputing from regional averages as the learning-poverty methodology does, produces a complete-looking matrix in which the poorest reporters have the least real data and the most invented data. For a diagnostic instrument that trade is unacceptable.

Statistical capacity is treated as a scored dimension within fiscal and administrative capacity rather than as a data problem to be worked around. A jurisdiction that cannot produce its own indicators has a capability deficit that the instrument should register, not conceal.

5.6 Robustness protocol

Five checks are mandatory before any result is reported, and all five are reported alongside it.

1. **Aggregation mode.** Report ranks under linear, geometric and penalised aggregation.
2. **Penalty strength.** Report the full path of ranks across gamma from 0 to 1.
3. **Goalpost choice.** Re-run with observed-sample goalposts and report rank movements.
4. **Indicator exclusion.** Drop each indicator in turn and report the maximum rank change, which is the standard leave-one-out sensitivity test.
5. **Data-quality weighting.** Re-run weighting each indicator by its assessed reliability, and report whether the ordering survives. If it does not, the ordering was an artefact of data quality and should not be published.

5.7 What the robustness protocol looks like on the demonstration data

The synthetic dataset of twelve archetypal profiles exists to show how much the ordering moves when the aggregation rule changes. The decisive case is the lopsided profile, which pairs strong delivery in five domains with a failed education system. Under linear aggregation it ranks third of twelve, ahead of the booking-centre economy and of a profile that is adequate everywhere and failing nowhere. Under the penalised rule it falls to fifth, below that same unremarkable profile. The two cross over between the rules, and the crossing is invisible in any single published ranking.

Archetype profile	GDP per capita index	Substance (linear)	Substance (geometric)	Substance (penalised, gamma=0.5)	Binding constraint
Mature welfare state	100	0.85	0.85	0.77	Work that sustains a life
Commodity exporter, strong institutions	88	0.71	0.71	0.58	Work that sustains a life
Booking-centre financial economy	145	0.61	0.61	0.46	Household material substance
Even but modest	38	0.54	0.54	0.42	Foundational competence

Archetype profile	GDP per capita index	Substance (linear)	Substance (geometric)	Substance (penalised, gamma=0.5)	Binding constraint
Lopsided: services without schooling	64	0.64	0.61	0.38	Foundational competence
Fast-growth assembler	72	0.51	0.51	0.35	Security against shocks
Learning-poor middle income	46	0.45	0.45	0.30	Foundational competence
Rent-financed high income	130	0.43	0.43	0.30	Work that sustains a life
Staged delivery state	40	0.40	0.40	0.27	Fiscal and administrative capacity
Frontier agrarian	14	0.38	0.38	0.25	Security against shocks
Remittance economy	34	0.38	0.38	0.25	Work that sustains a life
Fragile state	10	0.19	0.19	0.11	Security against shocks

Table 3. Twelve synthetic archetypal profiles under three aggregation rules. These are not countries and the values are constructed to illustrate behaviour, not to report findings. The final column is the diagnostic output the framework treats as its primary product.

Two features of the table carry the argument. First, the rent-financed high-income profile sits at the top of the GDP column and in the lower half of the penalised column, because its capability is purchased and its budget depends on a depleting rent. Second, the booking-centre profile has the highest measured output in the set and a household-material binding constraint, because the activity that inflates its output does not employ or house the people who live there.

5.8 The diagnostic outputs

The framework's product is four outputs per jurisdiction, not a rank.

- **The composite**, reported across the robustness path rather than at a point.
- **The binding constraint**: the domain with the lowest substance score.
- **Fragility flags**: domains where level is high and durability is low (staged), or level is high and endogeneity is low (purchased).
- **The gap between measured output rank and substance rank**, which is the single most policy-relevant number the instrument produces, because it locates the jurisdictions whose apparent success is least supported by their foundations.

5.9 A worked illustration of the reading

Take the staged-delivery profile in Table 3. Its domain scores are moderate to high on level, particularly in health and foundational competence, and its durability scores are the lowest in the set. The fragility flags fire on five of six domains, all in the staged direction: reported reach is well above the institutional base that would reproduce it.

The diagnosis this produces is specific. The problem is not that the health system fails to reach people; on the reported data it reaches more of them than a jurisdiction with twice its output per head. The problem is that the reach depends on a delivery mode rather than on a financing rule, a staffing establishment and a statutory entitlement, and that the three of those which would survive a change of government are absent. The policy implication is therefore not more delivery. It is conversion: turning reach into establishment, and campaign into statute.

Now take the purchased profile, which shows a different signature. Its level scores are high on the domains that money can buy directly, its endogeneity scores are low throughout, and its durability scores are low for the same reason. Here the flags fire in the purchased direction, and the diagnosis is about exposure rather than about institutions: the capability is real today and contingent on a price, a contract or a supply chain that the jurisdiction does not control.

The two profiles can carry the same composite score and require opposite policies. That is the argument for the diagnostic outputs over the composite, and it is why the composite should never be published without them.

6 Data critique

A framework of this kind is only as good as its refusal to over-claim. This section is a ledger: for each class of source, what it measures, how it can mislead, and what it cannot bear being used for.

6.1 Perception-based governance data

The Worldwide Governance Indicators aggregate perception sources through an unobserved components model. The method is well documented and the resulting series are widely used. What they measure is the perception of a jurisdiction's governance among respondents weighted towards commercial risk analysts, think tanks and survey respondents, not the direct observation of institutional behaviour. Two consequences follow. Margins of error are large enough that small rank differences are meaningless. And the sources are not independent of the country's commercial salience, which introduces a bias that runs in a consistent direction.

The framework therefore does not use perception indices as primary indicators. It uses observable administrative facts: tax collected, cases disposed, statistics produced, audits closed.

6.2 Modelled estimates

A large share of the social indicators used in any cross-country index are modelled rather than observed. The ILO's social protection coverage series is a modelled estimate built from the World Social Protection Database, national sources and extrapolation. The WHO's UHC service coverage index is a geometric mean of tracer indicators, several of which are themselves modelled. These are the best available estimates and they are produced by competent agencies; the point is that a modelled estimate carries uncertainty that a published point value conceals, and that the modelling assumptions are not uniform across countries.

The practical rule the framework adopts is that a modelled indicator may carry at most one-third of a domain's weight, and that any domain in which modelled indicators exceed that share is flagged in the output.

6.3 Learning assessment, and selection effects

Learning poverty combines a learning measure with an out-of-school adjustment. The learning measure draws on international, regional and national assessments in a documented hierarchy, and where none exists the value is imputed from the regional average. Participation in international assessments is voluntary, so the assessed set is not a random sample of the world. Meanwhile citizen-led surveys such as ASER reach rural districts only, which makes them excellent for tracking a rural trend and unusable as a national level estimate.

The magnitude is not in dispute even where the precision is. The 2022 update put learning poverty at 70% of 10-year-olds in low- and middle-income countries, against 57% before the pandemic, with South Asia at 78% and Sub-Saharan Africa at 89%. ASER 2023 found that among rural Indian youth aged 14 to 18, 43.3% could perform a three-digit by one-digit division and 26.5% could not read a Class 2 level text in their regional language. These are children who have been in school for eight to ten years.

6.4 Survey discontinuities read as trends

The most common analytical error in this field is not a bad indicator but a series break. India's labour force measurement moved from a quinquennial Employment-Unemployment Survey to an annual Periodic Labour Force Survey in 2017-18, and the treatment of unpaid helpers in household enterprises materially affects the measured female participation rate. Female participation on the usual status measure rose from 23.3% in 2017-18 to 41.7% in 2023-24. Some of that is real, some is measurement, and the decomposition is contested. Any framework that scores a trend must first establish that the series is continuous.

6.5 Reflexivity: the indicator degrades when targeted

Goodhart's observation and Campbell's law describe the same mechanism: a measure used for social decisions begins to be optimised directly, and stops measuring what it was chosen to measure. This is not a hypothetical risk for a diagnostic instrument. If a jurisdiction's allocation depends on its durability score, durability becomes something to be performed. The design response is to prefer indicators that are expensive to manipulate and to keep the underlying series visible rather than publishing only the composite.

6.6 Statistical capacity is a capability, not a nuisance

Countries reported at least one data point on 55% of the SDG indicators on average over 2015 to 2019; no country reported on more than 90%; 22 countries reported on under a quarter. The World Bank's Statistical Performance Indicators cover 51 indicators across 14 of 22 framework dimensions, with eight dimensions having no established measure at all. The capacity to measure capability is itself unequally distributed, and that inequality is correlated with the thing being measured.

There is a second-order version of the same point. Certain countries rely on donor funds and technical assistance to run their core household surveys. A jurisdiction whose statistics are produced by an external agency has an endogeneity problem in its measurement system that mirrors the endogeneity problem in its service delivery. The framework scores it rather than adjusting for it.

7 Reading the output: four archetypes

Plotting level against durability produces four positions, and each implies a different policy problem. This is the reading the instrument is built to support.

Position	Level	Durability	What it means	What follows
Solid	High	High	Capability is present and institutionally reproduced	Protect the institutional base; the risk is complacency and slow erosion
Staged	High	Low	Capability exists as delivery events rather than as a system	Convert reach into institutions: staffing, financing rules, statutory entitlement. The reported level overstates the stock
Purchased	High	Low (low endogeneity)	Capability is present but produced elsewhere or financed from a depleting source	Build domestic production of the capability, or accept a contingent position and price it
Latent	Low	High	Institutions and social reproduction are sound; delivery is weak	The cheapest position to improve: the base exists, the delivery does not
Hollow	Low	Low	Neither capability nor the institutions that would reproduce it	Sequencing problem: institution-building before delivery, or delivery will not persist

Table 4. Four positions on the level-durability plane, and the policy problem each implies.

The Kerala and Gujarat comparison is the clearest sub-national illustration of why the level-only reading is insufficient. Kerala's Human Development Index rose from 0.677 in 1999 to 0.790 in 2008; Gujarat's from 0.466 to 0.527; the all-India value from 0.387 to 0.467. Gujarat grew faster in aggregate output across the same period and converted less of it into human capability. The contested part of that literature is which state's trajectory is the better model; the uncontested part is that the two states' output rankings and capability rankings diverge, and that a framework which reports only one of them cannot describe either.

Kerala also illustrates why the framework needs the endogeneity axis rather than only level and durability. A substantial part of Kerala's household income has historically come from Gulf remittances, which is capability financed from outside the domestic production structure. That does not diminish the human development achievement, which is real and institutionally rooted. It does mean the profile is not the same shape as a jurisdiction that finances the same capability from domestic production, and the instrument should say so rather than average the difference away.

8 Limitations and falsifiability

A framework that cannot be wrong is not a framework. These are the conditions under which this one fails.

- **The endogeneity proxies are weak.** Import dependence, expatriate staffing and remittance shares are observable but crude, and a country can be deeply integrated into global production without being contingently capable. If the endogeneity axis turns out to correlate more with trade openness than with fragility, the axis is measuring the wrong thing and should be dropped.
- **Durability is inferred, not observed.** No cross-country series measures whether a capability survives a shock, because shocks are irregular. The proxies are structural features, and structural features are predictive rather than dispositive. The test is whether the durability score predicts actual capability loss in a subsequent crisis. That test has not been run here.
- **Equal domain weighting is arbitrary.** It is defensible on transparency grounds and it is still arbitrary. If the ordering is not robust to plausible reweighting, the instrument is producing a preference artefact.
- **The binding-constraint penalty embeds a value judgement** that a failed domain cannot be offset. That judgement is arguable and the gamma path exists so that a reader who disagrees can see what the disagreement costs.
- **It is a diagnostic, not a forecast.** Nothing here predicts growth, crisis or regime change, and any use of it in that mode would be unsupported.

9 Research agenda

Four pieces of work would turn this from a specification into an instrument.

1. **Construct the durability proxies as a validated scale.** This is the largest gap. It requires coding statutory versus discretionary entitlement across social programmes, measuring retention through at least three comparable fiscal consolidation episodes, and testing predictive validity against observed capability loss.
2. **Build the endogeneity measures from trade and fiscal microdata.** Concentration in the import basket for essential medicines, expatriate share of clinical staffing, remittance share of household consumption, and rent share of government revenue are all constructible from existing sources and none is currently assembled into a comparative series.
3. **Populate the indicator register for a pilot set of jurisdictions.** A pilot on fifteen to twenty jurisdictions, chosen to span the four archetypes rather than to maximise coverage, would establish whether the instrument discriminates in practice.
4. **Test the reflexivity risk.** Compare a jurisdiction's substance score against its subsequent behaviour once the score is public. If the score becomes a target, the instrument needs the redesign set out in section 6.5.

The framework is deliberately specified at a level of detail that makes it arguable. The domain boundaries, the indicator choices, the goalposts and the penalty strength are all judgement calls, stated so that they can be contested. That is the appropriate epistemic status for an instrument at this stage, and presenting it as anything more settled would be the failure mode this paper is arguing against.

Appendix A Indicator register

The register below lists the primary indicator proposed for each axis in each domain, the source that produces it, and the defect that limits its use. The full operational register is longer; this is the core set that determines the domain scores.

Domain	Axis	Primary indicator	Source	Principal defect
Foundational competence	L	Learning poverty: share of 10-year-olds below minimum reading proficiency	World Bank / UNESCO Institute for Statistics	Regional imputation for non-reporting countries
Foundational competence	L	Functional numeracy on an applied task, age 15	ASER; PISA; SEA-PLM	Voluntary participation; rural-only in citizen-led surveys
Foundational competence	E	Share of teachers trained in domestically financed institutions	National administrative data	Rarely published; not comparable across jurisdictions
Foundational competence	D	Learning retained after a six-month interruption	Ad hoc; pandemic-era studies	No regular series exists
Health capability	L	UHC service coverage index	WHO / World Bank	Modelled geometric mean of tracer indicators
Health capability	L	Catastrophic out-of-pocket spending above 10% of household budget	WHO Global Health Expenditure Database	Survey-based; understates forgone care
Health capability	E	Import dependence of essential medicines and diagnostics	National trade and pharma data	Aggregation across product classes is contested
Health capability	D	Statutory versus scheme-based entitlement to coverage	Legislative coding	Requires legal coding exercise; not yet standardised
Work that sustains a life	L	Informal employment share of total employment	ILO	Modelled; definitional variation across countries
Work that sustains a life	L	Female labour force participation, age 15 and over	ILO; national labour force surveys	Sensitive to treatment of unpaid household helpers
Work that sustains a life	L	Unpaid care burden, minutes per day, non-SNA production	National time use surveys	Infrequent; not harmonised across countries
Work that sustains a life	E	Remittance share of household consumption	World Bank; national balance of payments	Informal remittance channels undercounted
Security against shocks	L	Effective social protection coverage, at least one benefit	ILO World Social Protection Report	Modelled; counts receipt, not adequacy
Security against shocks	L	Share able to raise emergency funds within 30 days	Global Findex	Triennial; self-reported
Security against shocks	E	Share of core transfers financed domestically and delivered by statute	National budget documents	Requires classification judgement
Fiscal and administrative capacity	L	Tax to GDP; direct tax share of total tax	IMF; national finance ministries	Comparability of direct/indirect classification varies
Fiscal and administrative capacity	L	Judicial pendency: stock of cases and median time to disposal	National judiciaries	Definitions of a pending case differ
Fiscal and administrative capacity	L	Statistical performance index	World Bank SPI	Covers 14 of 22 dimensions; eight unmeasured
Household material substance	L	Real household consumption per capita	National accounts; household surveys	Survey-to-accounts gaps rarely reconciled

Domain	Axis	Primary indicator	Source	Principal defect
Household material substance	L	Food share of household budget	Household consumption surveys	Infrequent; basket definitions differ
Household material substance	E	Profit-shifting-adjusted national income where the gap is material	National statistics offices; IMF	Only a few jurisdictions publish an adjustment

Table 5. Core indicator register. Sources are named, not linked; the register is a specification, not a dataset.

Appendix B Aggregation algebra and sensitivity

Normalisation. For an indicator x with goalposts (x_min, x_max), the normalised value is n = (x - x_min) / (x_max - x_min), with inverse indicators normalised as n = (x_max - x) / (x_max - x_min). Values are clipped to [0,1] and any clipping is reported.

Domain substance. S(d) = L(d)^alpha x E(d)^beta x D(d)^gamma with alpha = 0.50, beta = 0.25, gamma = 0.25 and alpha+beta+gamma = 1. The exponents are weights in the multiplicative sense: they set how fast substance decays as an axis falls towards zero, not how important the axis is in a linear trade-off.

Composite. SI = G x [(1 - kappa) + kappa x min_d S(d)], where G is the geometric mean of domain substance across populated domains and kappa is the penalty strength, default 0.5. The penalty is multiplicative so that it cannot push the score above the geometric mean, which would make the weakest domain a bonus.

Boundedness. Since G in [0,1] and the bracket in [1 - kappa, 1], SI in [0,1] and SI <= G for all kappa >= 0. Monotonicity in any single domain holds for kappa < 1; at kappa = 1 the composite is not strictly increasing in non-binding domains, which is the intended lexicographic behaviour.

Degenerate cases. If any domain substance is exactly zero, G is zero under the geometric mean and the composite is zero for all kappa. In practice a floor of 0.01 is applied to each domain substance before aggregation, and the floor is reported. Without a floor a single missing or zero-valued indicator would annihilate the composite, which is a data-availability artefact rather than a finding.

Appendix C Demonstration profile matrix

The matrix below is the full input to Table 3, reproduced so that the arithmetic in section 5.7 can be checked. Every value is synthetic. Each cell shows level, endogeneity and durability in that order, on a 0 to 1 scale.

Archetypal profile	Foundational competence	Health capability	Work that sustains a life	Security against shocks	Fiscal and administrative capacity	Household material substance
Mature welfare state	0.88 / 0.86 / 0.90	0.86 / 0.82 / 0.88	0.80 / 0.84 / 0.82	0.86 / 0.88 / 0.90	0.84 / 0.86 / 0.86	0.82 / 0.86 / 0.84
Rent-financed high income	0.62 / 0.28 / 0.34	0.72 / 0.26 / 0.32	0.48 / 0.30 / 0.30	0.66 / 0.34 / 0.28	0.44 / 0.40 / 0.36	0.74 / 0.24 / 0.30
Fast-growth assembler	0.54 / 0.58 / 0.56	0.52 / 0.54 / 0.50	0.66 / 0.48 / 0.44	0.34 / 0.52 / 0.40	0.46 / 0.60 / 0.52	0.58 / 0.52 / 0.46
Frontier agrarian	0.24 / 0.74 / 0.62	0.26 / 0.60 / 0.48	0.34 / 0.70 / 0.58	0.18 / 0.66 / 0.52	0.22 / 0.72 / 0.60	0.20 / 0.68 / 0.54
Staged delivery state	0.58 / 0.46 / 0.24	0.62 / 0.40 / 0.20	0.42 / 0.48 / 0.30	0.50 / 0.42 / 0.22	0.38 / 0.44 / 0.26	0.46 / 0.50 / 0.28
Fragile state	0.16 / 0.30 / 0.22	0.18 / 0.28 / 0.20	0.22 / 0.32 / 0.24	0.10 / 0.26 / 0.18	0.12 / 0.24 / 0.16	0.14 / 0.30 / 0.20
Remittance economy	0.46 / 0.40 / 0.48	0.44 / 0.38 / 0.42	0.30 / 0.34 / 0.36	0.36 / 0.32 / 0.34	0.32 / 0.36 / 0.40	0.52 / 0.28 / 0.38

Archetypal profile	Foundational competence	Health capability	Work that sustains a life	Security against shocks	Fiscal and administrative capacity	Household material substance
Commodity exporter, strong institutions	0.74 / 0.78 / 0.80	0.72 / 0.74 / 0.76	0.64 / 0.62 / 0.66	0.70 / 0.76 / 0.74	0.68 / 0.72 / 0.70	0.70 / 0.70 / 0.68
Booking-centre financial economy	0.78 / 0.62 / 0.70	0.76 / 0.58 / 0.66	0.58 / 0.44 / 0.52	0.70 / 0.60 / 0.64	0.62 / 0.54 / 0.58	0.80 / 0.22 / 0.48
Learning-poor middle income	0.26 / 0.56 / 0.42	0.58 / 0.50 / 0.52	0.44 / 0.46 / 0.40	0.40 / 0.48 / 0.44	0.42 / 0.54 / 0.48	0.50 / 0.48 / 0.44
Lopsided: services without schooling	0.10 / 0.66 / 0.68	0.80 / 0.64 / 0.72	0.78 / 0.60 / 0.70	0.76 / 0.62 / 0.70	0.74 / 0.66 / 0.72	0.82 / 0.58 / 0.68
Even but modest	0.50 / 0.60 / 0.58	0.50 / 0.60 / 0.58	0.50 / 0.60 / 0.58	0.50 / 0.60 / 0.58	0.50 / 0.60 / 0.58	0.50 / 0.60 / 0.58

Table 6. Full synthetic input matrix. Cell order is Level / Endogeneity / Durability. These are constructed archetypes, not jurisdictions.

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